

MATHEMATICS FOR TECHNOLOGY



John Johnson
Cerritos College

Mathematics for Technology

This text is disseminated via the Open Education Resource (OER) LibreTexts Project (<https://LibreTexts.org>) and like the thousands of other texts available within this powerful platform, it is freely available for reading, printing, and "consuming."

The LibreTexts mission is to bring together students, faculty, and scholars in a collaborative effort to provide an accessible, and comprehensive platform that empowers our community to develop, curate, adapt, and adopt openly licensed resources and technologies; through these efforts we can reduce the financial burden born from traditional educational resource costs, ensuring education is more accessible for students and communities worldwide.

Most, but not all, pages in the library have licenses that may allow individuals to make changes, save, and print this book. Carefully consult the applicable license(s) before pursuing such effects. Instructors can adopt existing LibreTexts texts or Remix them to quickly build course-specific resources to meet the needs of their students. Unlike traditional textbooks, LibreTexts' web based origins allow powerful integration of advanced features and new technologies to support learning.



LibreTexts is the adaptable, user-friendly non-profit open education resource platform that educators trust for creating, customizing, and sharing accessible, interactive textbooks, adaptive homework, and ancillary materials. We collaborate with individuals and organizations to champion open education initiatives, support institutional publishing programs, drive curriculum development projects, and more.

The LibreTexts libraries are Powered by [NICE CXone Expert](#) and was supported by the Department of Education Open Textbook Pilot Project, the California Education Learning Lab, the UC Davis Office of the Provost, the UC Davis Library, the California State University Affordable Learning Solutions Program, and Merlot. This material is based upon work supported by the National Science Foundation under Grant No. 1246120, 1525057, and 1413739.

Any opinions, findings, and conclusions or recommendations expressed in this material are those of the author(s) and do not necessarily reflect the views of the National Science Foundation nor the US Department of Education.

Have questions or comments? For information about adoptions or adaptations contact info@LibreTexts.org or visit our main website at <https://LibreTexts.org>.

The LibreTexts name, logo, and book covers are trademarked by LibreTexts, Inc. (a 501(c)(3) not-for-profit organization) and protected against unauthorized use.

This text was compiled on 08/03/2026

- [InfoPage](#) by [Delmar Larsen](#) is licensed [Public Domain](#).

TABLE OF CONTENTS

Licensing

1: Module 1- Measurement and Geometry

- 1.1: Measurement
 - 1.1.1: Measurement and the United States System
 - 1.1.1.1: Exercises
 - 1.1.2: The Metric System of Measurement
 - 1.1.2.1: Exercises
 - 1.1.3: Conversion Between the Metric and US Customary Systems of Measurement
 - 1.1.3.1: Exercises
- 1.2: Geometry
 - 1.2.1: Perimeter and Area of Geometric Figures
 - 1.2.1.1: Exercises
 - 1.2.2: Circles
 - 1.2.2.1: Exercises
 - 1.2.3: Solid Geometric Figures and Objects
 - 1.2.3.1: Exercises
- 1.3: Percents
 - 1.3.1: Exercises
 - 1.3.2: Proportions and Rates
 - 1.3.3: Exercises

2: Module 2 - Finances

- 2.1: Taxes
 - 2.1.1: Income Taxation
 - 2.1.2: Exercises
- 2.2: Modeling with Linear Functions
 - 2.2E: Modeling with Linear Functions (Exercises)
- 2.3: Simple Interest
 - 2.3.1: Exercises
- 2.4: Modeling with Exponential Functions
 - 2.4E: Exponential Functions (Exercises)
- 2.5: Compound Interest
 - 2.5.1: Exercises
- 2.6: Annuities
 - 2.6.1: Exercises
- 2.7: Payout Annuities
 - 2.7.1: Exercises
- 2.8: Installment Loans
 - 2.8.1: Exercises

3: Module 3- Probability and Statistics

- 3.1: Basics of Sets
- 3.2: Union, Intersection, and Complement
- 3.3: Venn Diagrams
- 3.4: Introduction to Probability
 - 3.4.1: Exercises
- 3.5: Basic Concepts
 - 3.5.1: Exercises
- 3.6: Working with Events
 - 3.6.1: Exercises
- 3.7: Two-Way Tables
 - 3.7.1: Exercises
- 3.8: Counting
 - 3.8.1: Exercises
- 3.9: What is Statistics?
 - 3.9.1: Exercises
- 3.10: Sampling methods
 - 3.10.1: Exercises
- 3.11: Presenting Data Graphically
 - 3.11.1: Exercises
- 3.12: Reading Graphs in Media
 - 3.12.1: Exercises
- 3.13: Correlation and Causation, Scatter Plots
 - 3.13.1: Exercises
- 3.14: Measures of Central Tendency and the Normal Distribution

[Index](#)

[Glossary](#)

[Detailed Licensing](#)

[Detailed Licensing](#)

Licensing

A detailed breakdown of this resource's licensing can be found in [Back Matter/Detailed Licensing](#).

CHAPTER OVERVIEW

1: Module 1- Measurement and Geometry

1.1: Measurement

1.1.1: Measurement and the United States System

1.1.1.1: Exercises

1.1.2: The Metric System of Measurement

1.1.2.1: Exercises

1.1.3: Conversion Between the Metric and US Customary Systems of Measurement

1.1.3.1: Exercises

1.2: Geometry

1.2.1: Perimeter and Area of Geometric Figures

1.2.1.1: Exercises

1.2.2: Circles

1.2.2.1: Exercises

1.2.3: Solid Geometric Figures and Objects

1.2.3.1: Exercises

1.3: Percents

1.3.1: Exercises

1.3.2: Proportions and Rates

1.3.3: Exercises

This page titled [1: Module 1- Measurement and Geometry](#) is shared under a [CC BY](#) license and was authored, remixed, and/or curated by [Leah Griffith, Veronica Holbrook, Johnny Johnson & Nancy Garcia](#).

1.1: Measurement

Measurement

There are two major systems of measurement in use today. They are the *United States system* and the *metric system*.

The **metric system of measurement** is the decimal system of measure developed in the 1790s in France. This system which is also referred to as SI is the official unit used by every country except the United States. These units use a standard set of prefixes such as milli, centi, kilo, deci

The **US Customary Units** is a system of measurement primarily used in the United States which was validated in 1832. This system of measurement came from the English units, the units of measure used in the British Empire before the formulation of the Imperial system on January 1, 1826, and before American Independence. The customary units are widely used in the areas of commerce and industry. Consumer products and industrial manufacturing materials are weighed or measured using the customary units as evident in their labels.

The common customary units of measure used are:

- inch, foot, yard, and mile for lengths
- acre for area
- cubic inch, cubic foot, and cubic yard for volume and capacity
- ounce, pint, quart, and gallon for fluid volume
- pint and quart for dry volume

This section begins by defining measurement and its common uses.

This page titled [1.1: Measurement](#) is shared under a [not declared](#) license and was authored, remixed, and/or curated by [Leah Griffith](#), [Veronica Holbrook](#), [Johnny Johnson](#), and [Nancy Garcia](#).

1.1.1: Measurement and the United States System

6.1.1 Learning Objectives

- Define measurement.
- Perform arithmetic calculations on units of length, weight, and capacity.
- Convert from one unit of measure in the United States system to another unit of measure.
- Solve application problems using each form of measurement.

Measurement

Before we work the *United States measurement system* and the *metric system*, let's gain a clear understanding of the concept of measurement.

Definition: Measurement

Measurement is comparison to some standard.

Definition: Standard Unit of Measure

The concept of measurement is based on the idea of direct comparison. This means that measurement is the result of the comparison of two quantities. The quantity that is used for comparison is called the **standard unit of measure**.

Over the years, standards have changed. Quite some time in the past, the standard unit of measure was determined by a king. For example,

1 inch was the distance between the tip of the thumb and the knuckle of the king.

1 inch was also the length of 16 barley grains placed end to end.

Today, standard units of measure rarely change. Standard units of measure are the responsibility of the Bureau of Standards in Washington D.C.

Some desirable properties of a standard are the following:

1. *Accessibility*. We should have access to the standard so we can make comparisons.
2. *Invariance*. We should be confident that the standard is not subject to change.
3. *Reproducibility*. We should be able to reproduce the standard so that measurements are convenient and accessible to many people.

Definition: Length, Weight, Capacity

Length is a linear measurement to describe the distance between two objects.

Weight is used to describe how heavy or light an object is.

Capacity is used to describe the amount of liquid (or other pourable substance) that will fill an object.

The United States System of Measurement

Some of the common units (along with their abbreviations) for the United States system of measurement are listed in the following table.

Unit Conversion Table	
Length	1 foot (ft) = 12 inches (in)
	1 yard (yd) = 3 feet (ft)
	1 mile (mi) = 5,280 feet

Weight	1 pound (lb) = 16 ounces (oz) 1 ton (T) = 2,000 pounds
Liquid Capacity	1 tablespoon (tbsp) = 3 teaspoons (tsp) 1 fluid ounce (fl oz) = 2 tablespoons 1 cup (c) = 8 fluid ounces 1 pint (pt) = 2 cups 1 quart (qt) = 2 pints 1 gallon (gal) = 4 quarts

Conversions in the United States System

It is often convenient or necessary to convert from one unit of measure to another. For example, it may be convenient to convert a measurement of length that is given in feet to one that is given in inches. Such conversions can be made using *unit fractions*.

Definition: Unit Fraction

A **unit fraction** is a fraction with a value of 1 in the numerator or the denominator.

Unit fractions are formed by using two equal measurements. One measurement is placed in the numerator of the fraction, and the other in the denominator. **Placement depends on the desired conversion.**

Placement of Units

Place the unit being converted *to* in the **numerator**.

Place the unit being converted *from* in the **denominator**.

For example,

Equal Measurements	Unit Fraction
1 ft = 12 in	$\frac{1 \text{ ft}}{12 \text{ in}}$ or $\frac{12 \text{ in}}{1 \text{ ft}}$
1 pt = 16 fl oz	$\frac{1 \text{ pt}}{16 \text{ fl oz}}$ or $\frac{16 \text{ fl oz}}{1 \text{ pt}}$
1 wk = 7 da	$\frac{7 \text{ da}}{1 \text{ wk}}$ or $\frac{1 \text{ wk}}{7 \text{ da}}$

Example 1

Make the following conversions. If a fraction occurs, convert it to a decimal rounded to two decimal places.

Convert 11 yards to feet.

Solution

Looking in the unit conversion table under *length*, we see that $1 \text{ yd} = 3 \text{ ft}$. There are two corresponding unit fractions, $\frac{1 \text{ yd}}{3 \text{ ft}}$ and $\frac{3 \text{ ft}}{1 \text{ yd}}$. Which one should we use? Look to see which unit we wish to convert to. Choose the unit fraction with this unit in the *numerator*. We will choose $\frac{3 \text{ ft}}{1 \text{ yd}}$ since this unit fraction has feet in the numerator. Now, multiply 11 yd by the unit fraction. Notice that since the unit fraction has the value of 1, multiplying by it does not change the value of 11 yd.

$$\begin{aligned}
 11 \text{ yd} &= \frac{11 \text{ yd}}{1} \cdot \frac{3 \text{ ft}}{1 \text{ yd}} && \text{Divide out common units.} \\
 &= \frac{11 \cancel{\text{ yd}}}{1} \cdot \frac{3 \text{ ft}}{1 \cancel{\text{ yd}}} && (\text{Units can be added, subtracted, multiplied, and divided, just as numbers can.}) \\
 &= \frac{11 \cdot 3 \text{ ft}}{1} \\
 &= 33 \text{ ft}
 \end{aligned}$$

Thus, $11 \text{ yd} = 33 \text{ ft}$.

Example 2

Convert 36 fl oz to pints.

Solution

Looking in the unit conversion table under *liquid volume*, we see that 1 pt = 16 fl oz. Since we are to convert to pints, we will construct a unit fraction with pints in the numerator.

$$\begin{aligned}
 36 \text{ fl oz} &= \frac{36 \text{ fl oz}}{1} \cdot \frac{1 \text{ pt}}{16 \text{ fl oz}} && \text{Divide out common units.} \\
 &= \frac{36 \cancel{\text{ fl oz}}}{1} \cdot \frac{1 \text{ pt}}{16 \cancel{\text{ fl oz}}} \\
 &= \frac{36 \cdot 1 \text{ pt}}{16} \\
 &= \frac{36 \text{ pt}}{16} && \text{Reduce} \\
 &= \frac{9}{4} \text{ pt} && \text{Convert to decimals: } \frac{9}{4} = 2.25
 \end{aligned}$$

Thus, 36 fl oz = 2.25 pt.

Example 5

You want to lay out a red carpet walkway for your grandma's 75th birthday. The distance from the dropoff area to the entrance of the party venue is 18 feet. The fabric you want is only sold in yards. How many yards of fabric do you need?

Solution

The unit fraction needed for converting from feet to yards is $\frac{1 \text{ yd}}{3 \text{ ft}}$.

$$\begin{aligned}
 18 \text{ ft} &= \frac{18 \text{ ft}}{1} \cdot \frac{1 \text{ yd}}{3 \text{ ft}} && \text{Divide out common units.} \\
 &= \frac{18 \cancel{\text{ ft}}}{1} \cdot \frac{1 \text{ yd}}{3 \cancel{\text{ ft}}} \\
 &= \frac{18 \text{ yd}}{1 \cdot 3} \\
 &= 6 \text{ yd}
 \end{aligned}$$

Thus, 18 ft = 6 yd. You would need to purchase 6 yards of fabric for the red carpet.

Try It Now 1

Make the following conversions. If a fraction occurs, convert it to a decimal rounded to two decimal places.

1. Convert 2 mi to feet.
2. Convert 26 ft to yards.
3. Convert 9 qt to pints.

Answer

1. 10,560 ft
2. 8.67 yd.
3. 18 pt